Cyclistic Bike-Share Analysis

Understanding Casual Riders and Annual Members

October 2025 – September 2026

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Google Data Analytics Professional Certificate

Tools: Google BigQuery, SQL, Google Sheets

Executive Summary

Cyclistic is a bike-share company based in Chicago that aims to increase annual memberships by converting casual riders into members.

This case study analyzes 12 months of historical bike-share data, from October 2025 through September 2026, to understand how casual riders and annual members use Cyclistic bikes differently.

Key findings:

- Annual members account for approximately 65% of total rides, while casual riders account for 35%.
- Casual riders take longer trips, averaging 18.16 minutes compared with 12.17 minutes for members.
- Casual ridership is strongest on weekends, particularly Saturdays.
- Annual members demonstrate stronger weekday activity, with noticeable peaks during morning and evening commuting hours.
- Both rider groups show increased activity during summer months.

Recommendations:

Based on these findings, Cyclistic should focus on seasonal membership campaigns, targeted promotions for frequent weekend riders, and marketing that emphasizes the benefits of annual memberships for everyday transportation.

These strategies could help Cyclistic reach casual riders whose riding habits suggest potential interest in annual memberships.

Ask: Define the Business Task

Business Problem

Cyclistic wants to increase the number of annual memberships because annual members are more profitable than casual riders. The marketing team needs to better understand how these two customer groups use Cyclistic bikes to develop effective conversion strategies.

Business Objective

Analyze historical bike-share data from October 2025 through September 2026 to identify differences in riding behavior between casual riders and annual members.

The findings will help Cyclistic develop data-driven marketing recommendations aimed at converting casual riders into annual members.

Key Stakeholders

- **Lily Moreno:** Director of Marketing, responsible for developing strategies to increase annual memberships.
- **Cyclistic Marketing Analytics Team:** Responsible for analyzing rider behavior and supporting marketing decisions.
- **Cyclistic Executive Team:** Responsible for evaluating and approving the proposed marketing strategies.

Primary Business Question

How do annual members and casual riders use Cyclistic bikes differently?

Prepare: Data Sources and Organization

Data Source

The dataset consists of historical bike-trip records from Divvy, Chicago's bike-share system, made publicly available by Motivate International Inc.

The data is used for the fictional Cyclistic business case study as part of the Google Data Analytics Professional Certificate.

Data Collection Period

The analysis covers 12 months of bike-share activity, from October 1, 2025, through September 30, 2026.

Data Organization

The dataset contains individual trip records with information including:

- Ride ID and bicycle type
- Trip start and end timestamps
- Starting and ending stations
- Geographic coordinates
- Rider classification (casual or annual member)

The data was imported into Google BigQuery for cleaning, processing, and analysis.

Data Credibility and Limitations

The data originates from a publicly available bike-share dataset and provides a useful basis for identifying riding patterns.

However, the dataset does not contain personally identifiable information or detailed customer histories. Therefore, it cannot establish why individual riders use Cyclistic or whether a casual rider eventually purchases an annual membership.

The analysis focuses on observable riding behavior rather than assumptions about individual customers.

Process: Data Cleaning and Transformation

Tools Used

Google BigQuery and SQL were used to combine, clean, validate, and analyze the Cyclistic bike-share dataset.

Data Cleaning

The 12 monthly CSV files were imported into BigQuery and combined into a single table using `UNION ALL`.

The combined dataset contained 6,171,785 ride records.

Several data-quality checks were performed to identify duplicate ride IDs, missing values, invalid ride durations, and records outside the selected analysis period.

The following cleaning procedures were applied:

1. **Duplicate removal:** Used SQL's `ROW_NUMBER()` function to retain one record per unique ride ID.
2. **Ride duration validation:** Excluded rides shorter than one minute or longer than 24 hours to reduce the influence of questionable trip durations.
3. **Date filtering:** Restricted the dataset to rides beginning between October 1, 2025, and September 30, 2026.
4. **Missing values:** Confirmed that essential fields, including ride IDs, timestamps, and membership classifications, contained no missing values. Records with missing station names were retained because station information was not required for the primary analysis.

Data Transformation

Additional columns were created using SQL:

- `ride_length_seconds`: Duration of each ride in seconds.
- `ride_length_minutes`: Duration of each ride in minutes.
- `day_of_week`: Day of the week when each ride began.
- `ride_month`: Month when each ride began.

The original trip timestamps were preserved for calendar-based analysis after checking their representation against the source CSV files.

Final Dataset

After cleaning, the dataset contained **6,007,154 unique ride records**, representing approximately 97.33% of the original dataset.

The cleaned data was stored in a separate BigQuery table named `trips_cleaned`, preserving the original combined dataset for reference.

The cleaned dataset was then used to compare ride frequency, ride duration, weekday patterns, seasonal trends, and hourly riding behavior between casual riders and annual members.

Analyze: Identifying Ridership Patterns

Analysis Overview

Using SQL in Google BigQuery, I analyzed 6,007,154 cleaned ride records to identify differences in how casual riders and annual members use Cyclistic bikes.

The analysis focused on five areas: total ridership, average ride duration, weekly riding patterns, monthly trends, and hourly activity.

1. Total Ridership

Annual members accounted for 3,896,850 rides (64.9%), while casual riders accounted for 2,110,304 rides (35.1%).

This indicates that annual members represent the majority of Cyclistic's recorded trips.

2. Average Ride Duration

Casual riders averaged 18.16 minutes per ride, compared with 12.17 minutes for annual members.

Casual rides were approximately 49% longer on average, suggesting differences in how the two groups use the service.

3. Weekly Riding Patterns

Casual ridership was highest on Saturdays, with 447,265 rides. Annual members demonstrated stronger weekday activity, particularly on Wednesdays, with 631,330 rides.

These patterns suggest that casual riders may use Cyclistic more for weekend recreation, while annual members may rely on the service for routine transportation.

4. Monthly Ridership Trends

Both rider groups experienced increased activity during warmer months.

Casual ridership peaked in July 2026 with 346,230 rides, while annual membership ridership peaked in August 2026 with 510,566 rides.

These findings suggest that seasonal demand should be considered when developing marketing campaigns.

5. Hourly Riding Patterns

Annual members showed pronounced activity during morning and evening hours, particularly around 8 AM and 5 PM.

Casual ridership increased throughout the afternoon and peaked around 5 PM.

The stronger morning peak among annual members is consistent with commuting behavior, although trip purpose cannot be confirmed from the available data.

Overall Findings

The analysis revealed distinct differences between casual riders and annual members.

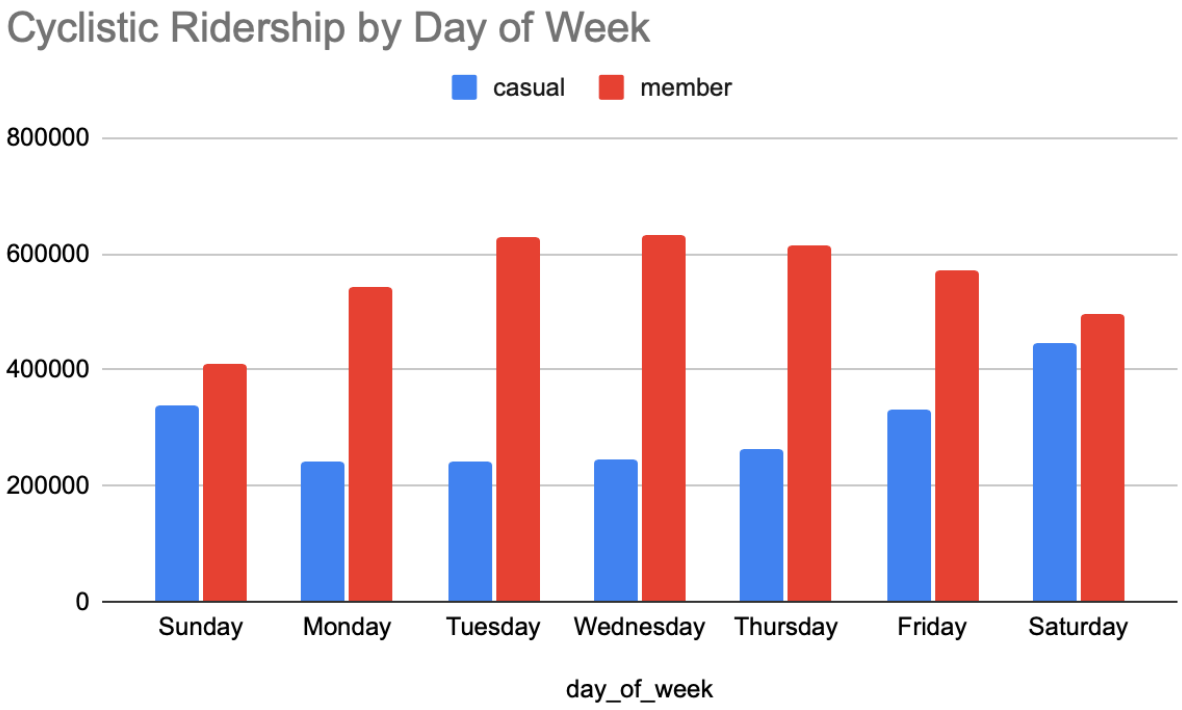
Casual riders generally take longer trips, ride more frequently on weekends, and demonstrate strong seasonal activity. Annual members account for most trips and show more consistent weekday usage, with riding patterns that suggest routine transportation.

These findings provide a foundation for developing targeted marketing strategies aimed at converting casual riders into annual members.

Share: Data Visualizations

To communicate the findings, four charts were created in Google Sheets using the results of SQL queries performed in BigQuery.

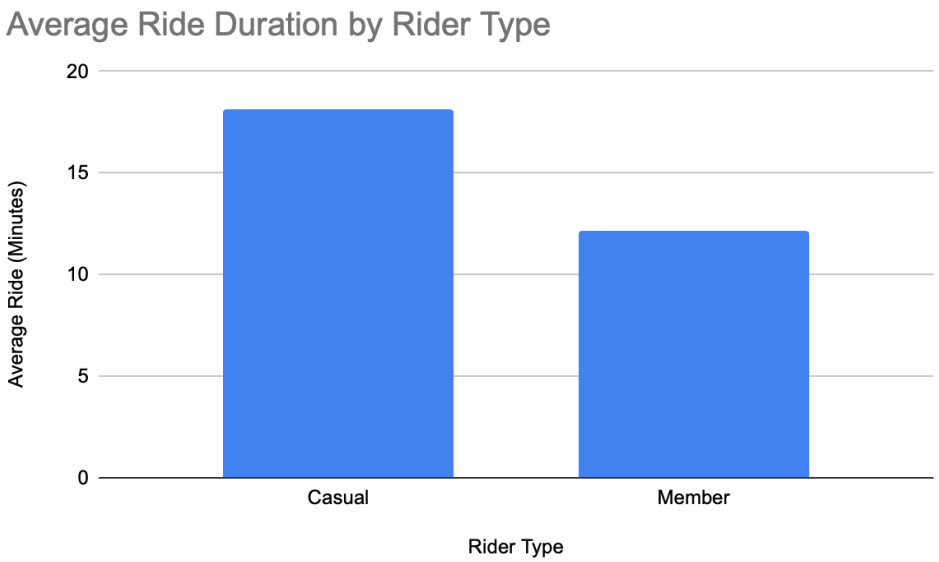
1. Weekly Ridership by Rider Type



Casual riders demonstrated their highest activity on Saturdays, while annual members recorded stronger ridership during weekdays.

This suggests that the two groups may have different transportation needs and riding habits.

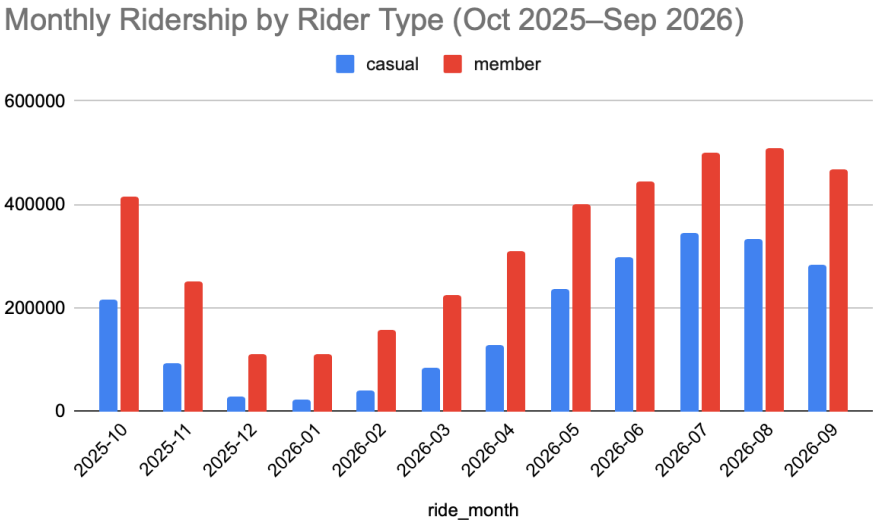
2. Average Ride Duration by Rider Type



Casual riders averaged 18.16 minutes per trip, compared with 12.17 minutes for annual members.

The longer average duration among casual riders may indicate greater recreational usage, although individual trip purposes cannot be confirmed.

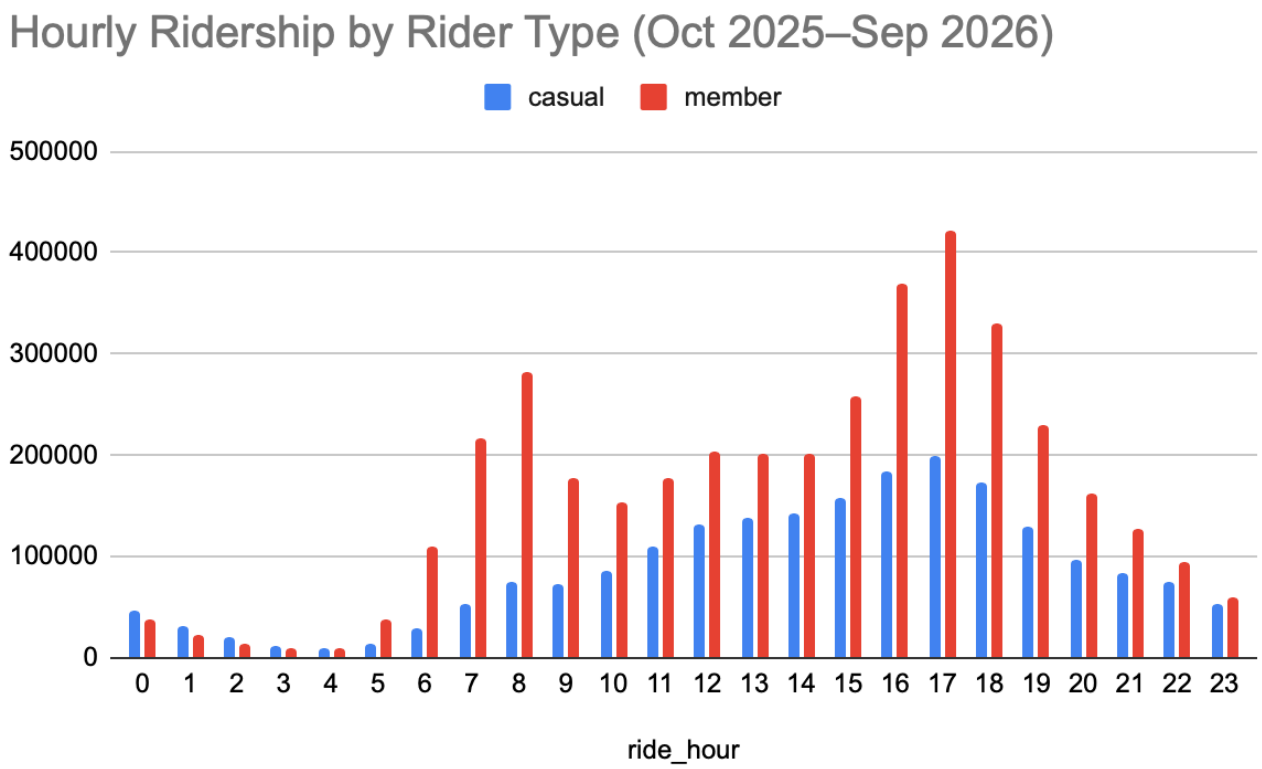
3. Monthly Ridership by Rider Type



Both groups experienced increased ridership during warmer months, with casual riders peaking in July and annual members peaking in August.

This highlights an opportunity to promote annual memberships during periods of increased casual riding activity.

4. Hourly Ridership by Rider Type



Annual members demonstrated noticeable morning and evening peaks, particularly around 8 AM and 5 PM.

Casual ridership increased throughout the afternoon, suggesting a different pattern of bike usage compared with annual members.

Visualization Summary

Together, these visualizations demonstrate meaningful differences in riding behavior between casual riders and annual members. These findings support marketing strategies focused on seasonal demand, weekend ridership, and everyday transportation.

Act: Recommendations and Conclusion

Recommendations

Based on the analysis of Cyclistic's historical bike-share data, I recommend three marketing strategies to encourage casual riders to become annual members.

1. Launch Seasonal Membership Campaigns

Casual ridership increases significantly during warmer months, particularly in July.

Cyclistic should introduce targeted membership promotions before and during the summer season to reach casual riders when demand is highest.

2. Target Weekend Riders

Casual riders take longer trips and demonstrate their highest ridership on Saturdays.

Cyclistic should develop marketing campaigns highlighting the potential benefits of annual memberships for weekend and recreational riding. These campaigns could be promoted through the Cyclistic app and at popular bike stations.

3. Promote Everyday Transportation

Annual members demonstrate stronger weekday ridership, with noticeable activity peaks during morning and evening commuting hours.

Cyclistic should emphasize the convenience of annual memberships for everyday transportation, encouraging casual riders to consider using bicycles beyond recreational activities.

Conclusion

This analysis identified meaningful differences between casual riders and annual members across more than six million bike-share trips.

Casual riders generally take longer trips and show stronger weekend and seasonal activity, while annual members demonstrate more consistent weekday usage and commuting-like patterns.

These findings suggest opportunities for Cyclistic to tailor its marketing efforts toward casual riders based on when and how they use the service.

Although the available data cannot establish individual rider motivations or predict membership conversions, the analysis provides a foundation for testing targeted marketing strategies.

By evaluating the effectiveness of these campaigns, Cyclistic can make more informed decisions about increasing annual memberships.

Appendix A: SQL Methodology

Tools and Environment

- **Database:** Google BigQuery
- **Language:** SQL (GoogleSQL)
- **Visualization:** Google Sheets
- **Dataset:** `cylistic-project-one.cyclistic`
- **Cleaned table:** `cylistic-project-one.cyclistic.trips_cleaned`

Data Preparation

Twelve months of historical bike-share data were combined and cleaned using SQL.

The cleaning process included removing duplicate ride IDs, excluding invalid ride durations, filtering records to the analysis period, and creating additional columns for ride duration and calendar-based analysis.

Analytical Queries

SQL queries were used to examine:

1. Total rides by membership type
2. Average ride duration by membership type
3. Ridership by day of the week
4. Monthly ridership trends
5. Ridership by hour of the day

SQL Repository

The SQL scripts used for data cleaning and analysis are available here:

https://github.com/kenyanamerican/cyclistic-bike-share-analysis

Reproducibility

The SQL scripts document the steps used to prepare and analyze the dataset. The resulting aggregated data was exported to Google Sheets to create the visualizations presented in this report.